PROJECT OUTCOMES AND JUSTIFICATION

Faculty Name: Mr. Gnanesh Academic Year: 2025-2026
Branch: CSE Year/Sem: IV/II

P-PO Mapping Justification

Mapping	Level	Justification
The above mapping is justified as below by students		
P1-PO1	H	Apply engineering and AI/ML techniques to build an integrated mental health assessment system for emotional analysis and personalized feedback.
P1-PO2	M	Use research and first principles to analyze mental health challenges and develop AI models for mood detection, stress analysis, and behavioral insights.
P1-PO3	H	Design an intelligent mental health system addressing user well-being with societal and ethical considerations.
P1-PO5	H	Address mental health challenges using data analysis and AI models to build a unified platform for emotion detection and support.
P1-PO7	H	Integrated ML models into a unified system to tackle agricultural challenges, considering environmental impact and promoting sustainability.
P1-PO8	M	Ensure ethical responsibility in handling sensitive mental health data and AI-based decision-making.
P1-PO9	H	Work collaboratively in a multidisciplinary team to develop AI models for emotion detection, voice analysis, and user behavior tracking.
P1-PO10	H	Communicate complex system functionalities effectively through reports, documentation, and presentations.
P1-PO11	H	Apply engineering and management principles in developing and integrating AI-based mental health solutions.

P1-PO12	M	Recognize the importance of lifelong learning in AI and mental health technologies for continuous improvement.
P2-PO1	M	Analyze mental health requirements and design a user-friendly platform providing actionable insights.
P2-PO3	H	Develop a system offering real-time insights using first principles and user-centered design.

P2-PO5	H	Apply modern tools and technologies to develop a responsive and interactive mental health platform.
P2-PO8	H	Ensure ethical standards and privacy protection in building the mental health system.
P2-PO9	H	Collaborate effectively in teams to develop a comprehensive mental health web application.
P2-PO10	H	Present mental health insights clearly through dashboards, reports, and UI interactions.
P2-PO11	H	Apply project management skills in developing and maintaining the platform.
P2-PO12	M	Adapt to evolving technologies in AI and healthcare applications.
P3-PO1	H	Apply AI/ML knowledge to provide data-driven mental health insights.
P3-PO2	H	Analyze complex behavioral and emotional patterns using research-based approaches.
P3-PO3	H	Design solutions addressing mental health, societal well-being, and user engagement.
P3-PO5	M	Use modern tools to deliver accurate predictions and personalized recommendations.
P3-PO6	H	Consider user safety, mental well-being, and ethical aspects in system design.
P3-PO7	M	Promote mental wellness and healthy lifestyle practices through digital tools.
P3-PO8	M	Follow ethical guidelines in handling sensitive psychological data.
P3-PO9	H	Collaborated effectively within multidisciplinary teams.
P3-PO11	M	Apply project and team management skills to deliver application features.
P3-PO12	H	Continuously adapt to new AI technologies and mental health innovations.
P4-PO1	H	Apply engineering knowledge to design a scalable real-time mental health monitoring system.
P4-PO3	L	Design scalable systems considering user safety, privacy, and reliability.

P4-PO6	M	Address societal responsibility in building mental health support systems.
P4-PO7	H	Promote sustainable mental health practices using digital solutions.
P4-PO9	M	Collaborate to build a scalable and future-ready platform.
P4-PO10	H	Communicate system architecture and design effectively to stakeholders.
P4-PO11	M	Apply management principles to develop scalable application architecture.
P4-P012	H	Demonstrate adaptability and continuous learning in emerging health technologies.

P-PSO Mapping Justification

Mapping	Level	Justification
Students will be able to		
P1-PSO1	H	Design an AI-based mental health system addressing societal well-being.
P1-PSO2	M	Apply machine learning techniques for emotion detection and behavioral analysis.
P2-PSO1	H	Develop a web-based platform providing mental health support and insights.
P2-PSO2	H	Utilize full-stack development and AI integration for healthcare applications.
P3-PSO1	H	Deliver data-driven insights to improve user mental health outcomes.
P3-PSO2	H	Apply AI and data science for real-time emotional and behavioral analysis.
P4-PSO1	M	Enable future-ready digital mental health systems for societal benefit.
P4-PSO2	L	Build a scalable architecture supporting future AI and IoT integration.

P-PEO Mapping Justification

Mapping	Level	Justification
Students will be able to		
P1-PEO1	H	Apply computer engineering principles to solve real-world mental health problems using AI.
P1-PEO2	H	Use core engineering knowledge to develop intelligent healthcare applications.
P1-PEO3	M	Utilize emerging technologies such as AI, ML, and data science.
P1-PEO4	H	Develop scalable and future-ready healthcare systems promoting lifelong learning.
P2-PEO1	H	Demonstrate professional and analytical skills in application development.
P2-PEO2	H	Design and implement innovative real-world mental health solutions.
P2-PEO3	H	Use modern full-stack and AI tools effectively.
P2-PEO4	M	Foster teamwork and adaptability for evolving technologies.
P3-PEO1	H	Enhance analytical skills through meaningful mental health data insights.
P3-PEO2	M	Solve real-world problems using AI-based predictions.
P3-PEO3	H	Apply modern AI technologies in decision-making systems.

P3-PEO4	M	Encourage continuous improvement through evolving technologies.
P4-PEO1	M	Develop scalable engineering solutions for healthcare systems.
P4-PEO2	L	Build forward-compatible systems addressing real-world needs.
P4-PEO3	H	Integrate advanced technologies such as AI and cloud computing.
P4-PEO4	L	Promote lifelong learning through future-oriented system design.

DECLARATION

We hereby declare that the results embodied in the dissertation entitled **"EMBRACE MIND- A HOLISTIC MENTAL HALTH APP "** has been carried out by us together during the academic year 2025-26 as a partial fulfillment of the award of the B.Tech degree in Computer Science and Engineering from Keshav Memorial Institute of Technology (An Autonomous Institution) Narayanaguda, affiliated to JNTUH. We have not submitted this report to any other university or organization for the award of any other degree.

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ACKNOWLEDGEMENT

We take this opportunity to thank all the people who have rendered their full support to our project work. We render our thanks to **Dr. B L Malleswari**, Principal who encouraged us to do the Project.

We are grateful to **Mr. Neil Gogte**, Founder & Director, **Mr. S. Nitin,** Director for facilitating all the amenities required for carrying out this project.

We express our sincere gratitude to **Ms. Deepa Ganu**, Director Academic for providing an excellent environment in the college.

We are also thankful to **Mr. Para Upendar**, Head of the Department for providing us with time to make this project a success within the given schedule.

We are also thankful to our Faculty Supervisor **Mr. Gnanesh**, for her/his valuable guidance and encouragement given to us throughout the project work.

We would like to thank the entire CSE Department faculty, who helped us directly and indirectly in the completion of the project.

We sincerely thank our friends and family for their constant motivation during the project work.

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ABSTRACT

This project presents a comprehensive web-based mental health application designed to enhance emotional well-being through intelligent, interactive, and personalized support systems. Developed using React for the frontend and Flask for backend services, the platform integrates multiple wellness tools such as mood tracking, guided meditation, breathing exercises, gratitude journaling, goal management, and community interaction.

The application uses advanced emotion-recognition technologies, including an LSTM-based facial expression detection model and voice modulation analysis, to assess users' real-time emotional states. A dynamic mental level testing module generates adaptive questions, computes personalized scores, and visualizes progress through a mental health graph.

Additional features like VoChat for emotional assistance, one-to-one chat sessions, yoga guidance, and a stress-relief Bubble Pop game further improve engagement and support healthy habits. Overall, this system delivers a holistic, adaptive, and accessible environment that helps users understand their emotions, build resilience, and improve overall mental well-being.

Keywords : Mental Health, Emotion Detection, LSTM, Voice Modulation Analysis, Mood Tracking, Wellness Application, AI Chat Support, Web Application, Yoga Guidance, Stress Relief Game, Flask Backend, React Frontend, Mental Level Testing, User Well-Being.

Domain: Artificial Intelligence, Web Development, Mental Health & Wellness Technology, Human–Computer Interaction (HCI).

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CHAPTER-1

1. INTRODUCTION

1.1 Purpose of the project

The purpose of this project is to create an intelligent and interactive mental health application that helps users understand, track, and improve their emotional well-being. The platform combines traditional wellness practices with modern AI-driven technologies to offer personalized guidance, emotional support, and real-time mental state analysis. By integrating mood tracking, meditation, yoga, journaling, activity monitoring, and advanced emotion-recognition tools, the system aims to deliver a holistic environment that promotes mental stability, resilience, and positive behavioral change.

1.2 Problem with Existing Systems

Most existing mental health applications have limited functionality and fail to address user needs comprehensively. Common issues include:

- Lack of real-time emotional analysis: Many apps cannot detect mood changes through facial expressions or voice modulation.
- No personalized insights: Traditional apps provide generic suggestions rather than tailored recommendations.
- Fragmented tools: Users often need multiple apps for meditation, journaling, habit tracking, and emotional support, leading to inconsistent usage.
- Limited interactivity: Emotional support is usually static, with no conversational assistance or real-time engagement.
- Poor motivation mechanisms: Many platforms lack gamification or interactive features to keep users consistently engaged.

1.3 Proposed System

The proposed system is an all-in-one mental wellness platform that integrates advanced AI features with traditional mental health tools. Key components include:

- LSTM-based Face Emotion Detection to assess the user's emotional state through the "Vibe Check" feature.
- Voice Modulation Analysis to identify changes in tone, stress, or emotional variations.
- Mood Tracking and Visualization to monitor mental health trends over time.
- Meditation, Yoga Sessions, and Breathing Exercises for stress reduction and mental relaxation.
- Gratitude Journal and Goal Tracker to promote positivity and personal growth.
- Mental Level Testing with dynamically generated questions for personalized assessment.

- VoChat AI Support and One-to-One Chat System for emotional assistance and guidance.
- Community Support Section for engagement and motivation.
- Stress-relief Bubble Pop Game for fun and emotional balance.
- Admin Panel for managing users and monitoring system insights.

1.4 Scope of the Project

The scope of this project includes:

- Providing a web-based interface accessible from any device.
- Offering a comprehensive set of wellness tools including meditation, breathing practices, yoga, journaling, and goals.
- Implementing AI-driven emotional analysis through face and voice data.
- Delivering real-time insights and personalized recommendations for mental health improvement.
- Supporting secure authentication and user management, including an admin view.
- Enabling interactive features, such as chat support and gamified stress-relief activities.
- Visualizing user progress through day-by-day mental health graphs.
- Ensuring the system can be extended with more AI models, games, or wellness modules in the future.

1.5 Architecture Diagram

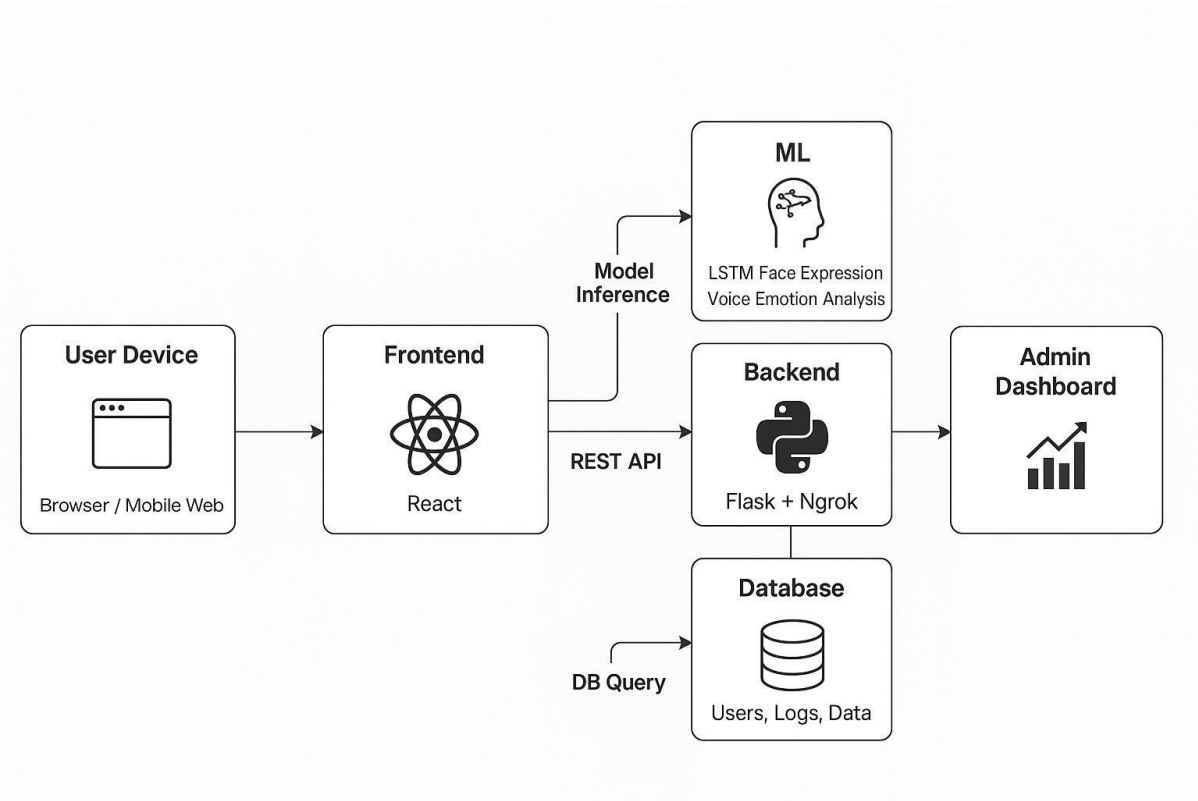

Fig-1 : Architecture Diagram

CHAPTER-2

2. LITERATURE SURVEY

1. Introduction & Rationale

Digital mental-health interventions (mobile/web apps and chatbots) have grown rapidly as scalable tools to complement traditional care because they can improve accessibility and deliver structured self-help interventions remotely. Recent systematic reviews and meta-analyses show that some app-based interventions produce small-to-moderate reductions in depression and anxiety symptoms, especially when they are evidence-based and personalized.

2. Effectiveness of Mental-Health Apps

Systematic reviews of smartphone and web apps indicate positive outcomes for anxiety, depression and specific phobias when interventions are structured (CBT-style content, reminders, interactive tasks) and used consistently. However, effect sizes vary, many trials are short-term, and quality across commercial apps is inconsistent—highlighting the need for validated, user-centered design and long-term follow-up.

3. AI Chatbots & Conversational Agents

AI chatbots (e.g., Woebot, Tess) have been trialed as scalable delivery platforms for CBT and supportive conversations. Randomized and quasi-experimental studies suggest chatbots can reduce symptoms in subclinical and mild-to-moderate cases when engagement is sustained .

4. Facial Emotion Recognition (FER) — Models & Trends

Facial-expression recognition is widely studied for emotion inference. Hybrid deep-learning approaches that combine convolutional neural networks (CNNs) for spatial feature extraction with recurrent architectures (LSTM) for temporal modelling of frame sequences (CNN–LSTM) have shown improved accuracy on dynamic expression datasets compared to static-frame methods. Recent comparative studies report competitive performance with attention or bidirectional LSTM enhancements.

5. Speech / Voice Emotion Recognition

Speech Emotion Recognition (SER) has matured with deep learning (CNN, LSTM, CNN–LSTM, attention-based models). Reviews show SER can discriminate emotional states and has been successfully applied to screen for mood disorders in controlled studies; however, speaker variability, language/accent differences, and noisy environments remain key challenges. Recent mental-health–focused reviews indicate voice features can be useful biomarkers for mood disorders when combined with other modalities.

8. Multi-modal Fusion — Best Practice

Recent literature increasingly supports multi-modal approaches (face + voice + self-report + behaviour logs) for more reliable emotion/mood inference than single-modality systems.

CHAPTER-3

3. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.1 Introduction to SRS

A Software Requirement Specification (SRS) is a comprehensive document that outlines the complete requirements of a software system. It clearly describes the purpose, functionalities, constraints, performance criteria, interfaces, and overall behavior of the application. An SRS ensures that developers, designers, testers, and stakeholders have a common understanding of the system before development begins. It acts as a blueprint that guides the entire software development lifecycle.

3.2 Role of SRS

The SRS plays several critical roles in software engineering:

- **Communication Tool:** Bridges the gap between stakeholders and the development team.
- **Reference Document:** Provides a single source of truth throughout development.
- **Foundation for Design & Implementation:** Guides developers in building system architecture and modules.
- **Basis for Testing:** Helps testers create test cases and verify system functionality.
- **Scope Management:** Prevents scope creep by clearly defining system boundaries.
- **Cost & Time Estimation:** Helps in planning resources, timelines, and budgets accurately.

3.3 Requirements Specification Document

The SRS document includes:

- **Detailed Functional Requirements:** What the system should do.
- **Non-Functional Requirements:** Qualities like performance, security, usability, reliability.
- **System Interfaces & Architecture:** How modules interact internally and externally.
- **Data Requirements:** Storage, retrieval, security, and processing constraints.
- **Hardware & Software Requirements:** Tools and environment needed to run the system.
- **Performance Expectations:** Response time, accuracy, scalability, and reliability metrics.

3.4 Functional Requirements

These describe **what** the system must perform.

1. User Authentication & Profile Management

- Users must be able to register, log in, update profile, and reset passwords.
- Admin must be able to manage users and system data.

2. Mood Tracking

- Users can log their daily mood.
- System displays mood history and visual graphs.

3. Face Emotion Detection (Vibe Check)

- Real-time facial emotion detection using the LSTM model.
- System identifies emotions such as happy, sad, angry, calm, etc.

4. Voice Modulation Analysis

- Detect emotional patterns through voice tone and speech.

5. Meditation, Breathing & Yoga Modules

- Users can access guided meditation and breathing exercises.
- Yoga sessions tailored to user needs.

6. Gratitude Journal & Goal Tracker

- Users can write daily gratitude notes.
- Users can set and track personal goals.

7. Mental Level Testing

- Dynamic question generation.
- The system calculates a mental score and displays a progress graph.

8. VoChat AI Support

- AI chatbot for emotional support and guidance.

9. One-to-One Chat Sessions

- Users can chat privately for personalized assistance.

10. Community Support Portal

- Users can interact, share thoughts, and motivate each other.

11. Stress Relief Bubble Pop Game

- Interactive game to reduce stress and improve focus.

12. Admin Dashboard

- Admin can manage users, track reports, and monitor system activity.

3.5 Non-Functional Requirements

1. Usability

- Interface should be simple, clean, and easy to navigate.
- Support for both light and dark modes.

2. Reliability

- Application should handle continuous usage without failures.
- Data should not be lost during operations.

3. Security

- Password encryption and secure API communication.
- Sensitive data (face/voice inputs) handled securely.

4. Scalability

- System should support increasing users without performance degradation.

5. Portability

- Application should run smoothly on all modern browsers and devices.

6. Maintainability

- Code should follow modular design for updates and extensions.

7. Availability

- System should strive for 99% uptime.

3.6 Performance Requirements

- Facial emotion detection should process input with minimal delay (1–3 seconds).
- Chat responses should appear in under 2 seconds.
- Dashboard graphs should load within 1 second.
- System must support **at least 500 concurrent users** in initial version.
- Voice modulation analysis should work efficiently without noticeable lag.
- Game interactions must respond instantly for smooth gameplay.

3.7 Software Requirements

Frontend

- React.js
- HTML5, CSS3, JavaScript
- Chart libraries for graphs (e.g., Chart.js / Recharts)

Backend

- Python Flask
- Ngrok (for secure tunneling and API exposure)
- REST API architecture

Machine Learning

- Python
- LSTM-based model with TensorFlow/Keras
- OpenCV for face detection
- Audio processing libraries (Librosa / PyAudio)

Database

- MySQL / PostgreSQL / MongoDB
- SQLAlchemy ORM (optional)

Tools

- VS Code
- Postman
- Git & GitHub

3.8 Hardware Requirements

For End Users

- A device with an internet-enabled browser (Laptop/PC/Tablet/Phone)
- Camera (for face emotion detection)
- Microphone (for voice analysis)
- Minimum 2GB RAM recommended

For Developers / Deployment Server

- Processor: Quad-core or better
- RAM: Minimum 8GB (recommended 16GB for ML processing)
- Storage: 40GB+
- GPU (optional but useful for ML training)
- Stable internet connection for Ngrok and server communication

CHAPTER-4

4. SYSTEM DESIGN

4.1 Introduction to UML

Unified Modeling Language (UML) is a standardized visual modeling language used to represent the structure, design, and behavior of a software system. It helps developers, designers, and stakeholders understand how different components of the system interact with each other. UML diagrams describe both **static** and **dynamic** aspects of the system, improving clarity, communication, and design accuracy before actual development begins.

UML is widely used during system analysis and design phases because it:

- Visualizes system architecture
- Identifies relationships between modules
- Communicates design decisions
- Helps detect design issues early
- Acts as a blueprint for implementation

4.2 UML Diagrams

For the mental health application, the following UML diagrams are used to illustrate the system's design:

- **Use Case Diagram** – Shows interactions between users and the system.
- **Sequence Diagram** – Shows how objects interact in time order.
- **State Chart Diagram** – Shows different states of the system or module.
- **Deployment Diagram** – Shows physical deployment of software components.

4.2.1 Use Case Diagram

Actors:

- User
- Admin
- AI Engine (ML Models)
- VoChat Chatbot

Major Use Cases:

- Register / Login
- Track Mood
- Use Vibe Check (Face Emotion Detection)
- Voice Modulation Analysis
- Access Meditation / Breathing / Yoga

- Write Gratitude Journal
- Play Bubble Pop Game
- Take Mental Level Test
- Chat with VoChat
- Use One-to-One Chat
- Manage Profile
- Admin Dashboard Management

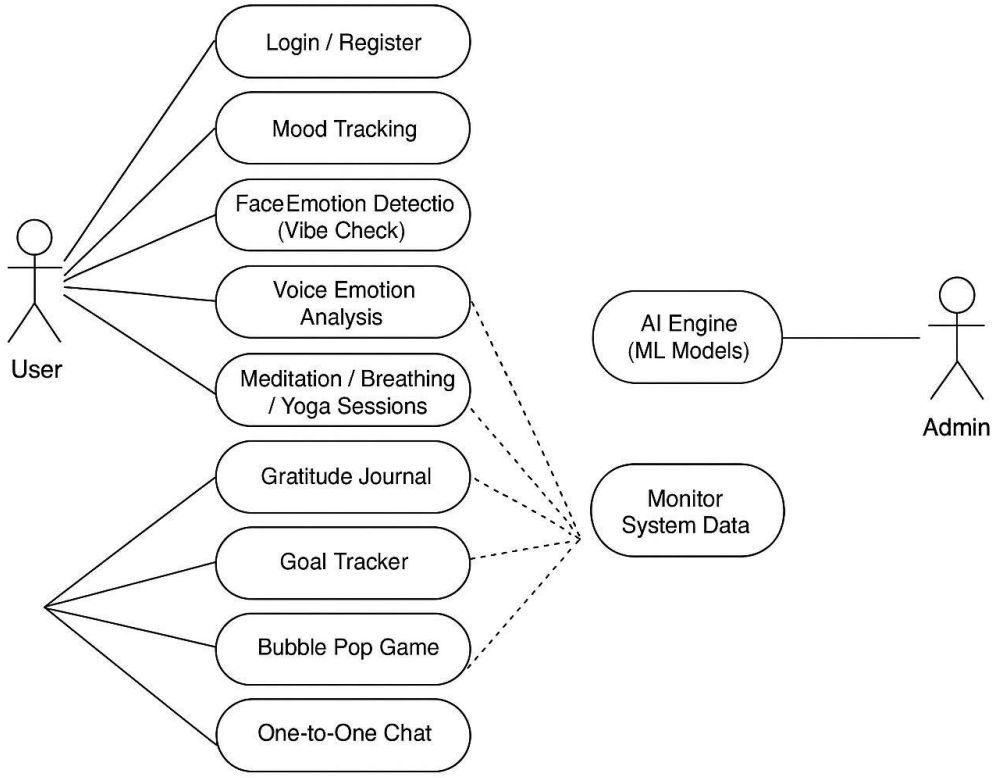

Fig-2: Usecase Diagram

4.2.2 Sequence Diagram

Sequence Diagram for “Face Emotion Detection (Vibe Check)”

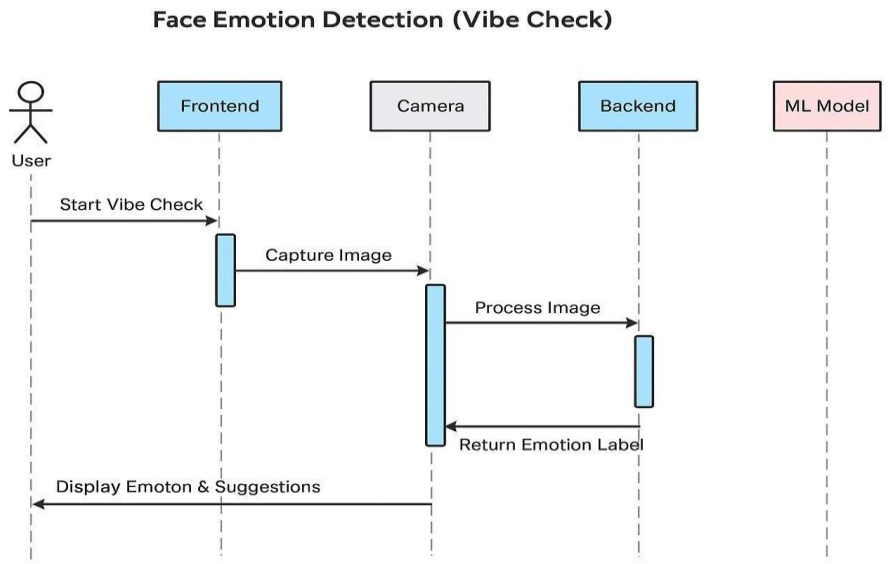

Fig-3: Sequence Diagram for Vice Check

Sequence Diagram for “VoChat AI Chat Support”

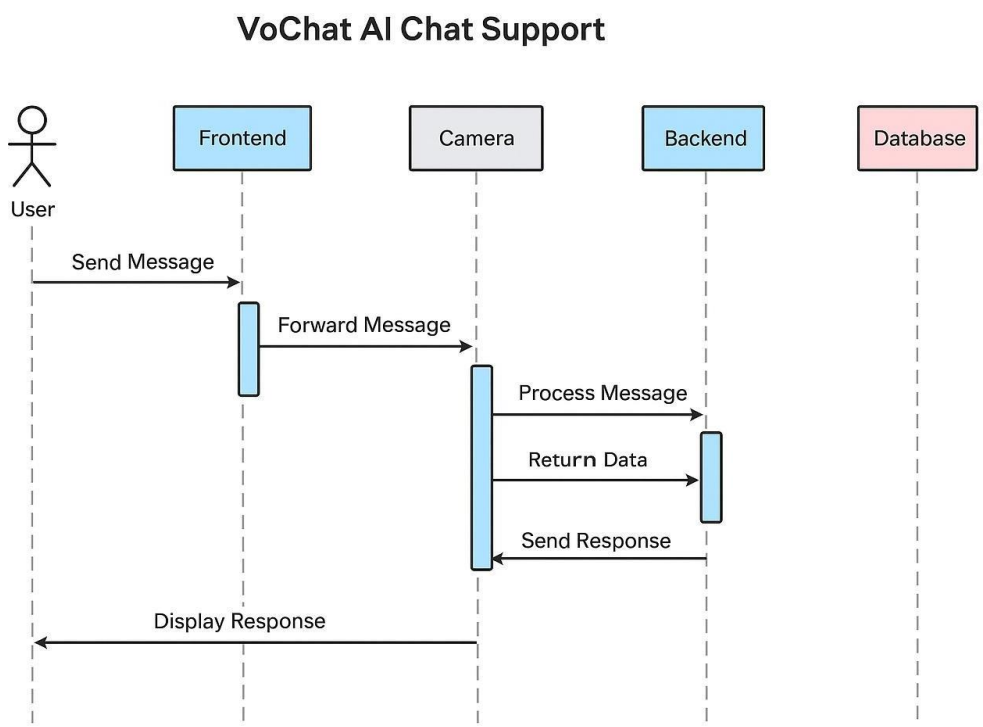

Fig-4: Sequence Diagram for Vice Check

4.2.3 State Chart Diagram

State Diagram for a User Session

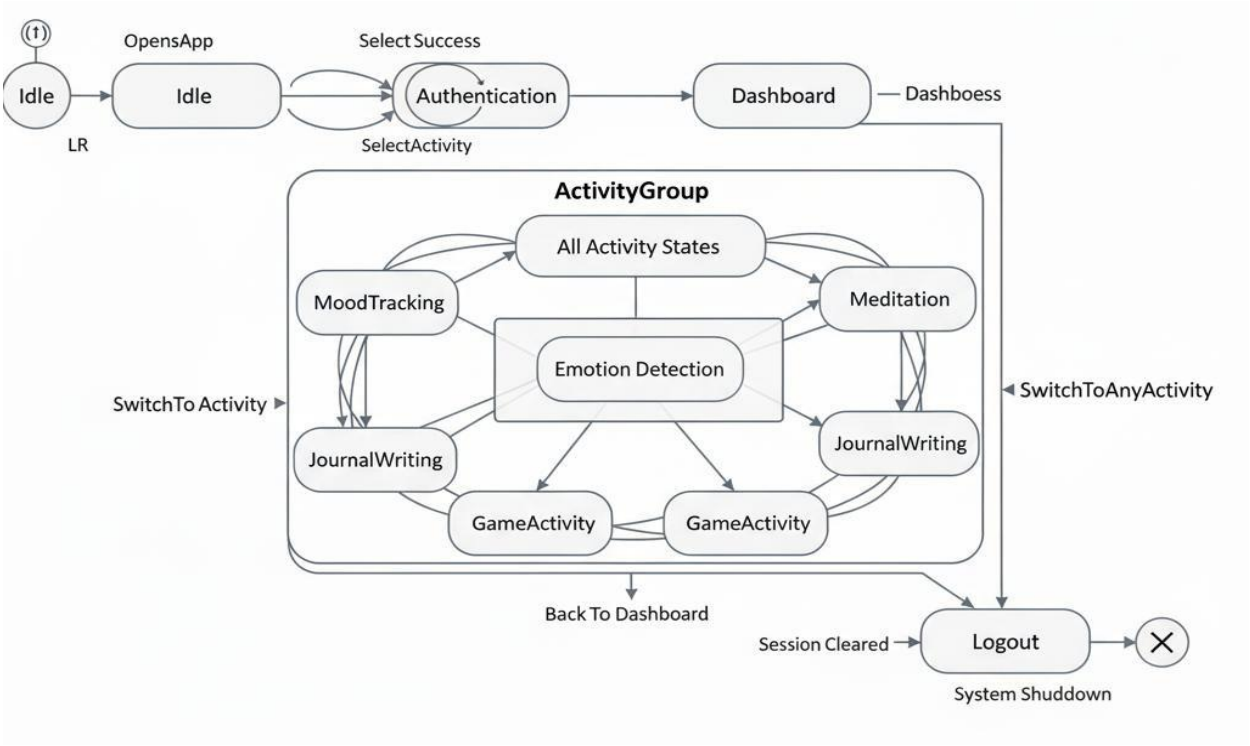

Fig-5: State Diagram for user session

4.2.4 Deployment Diagram

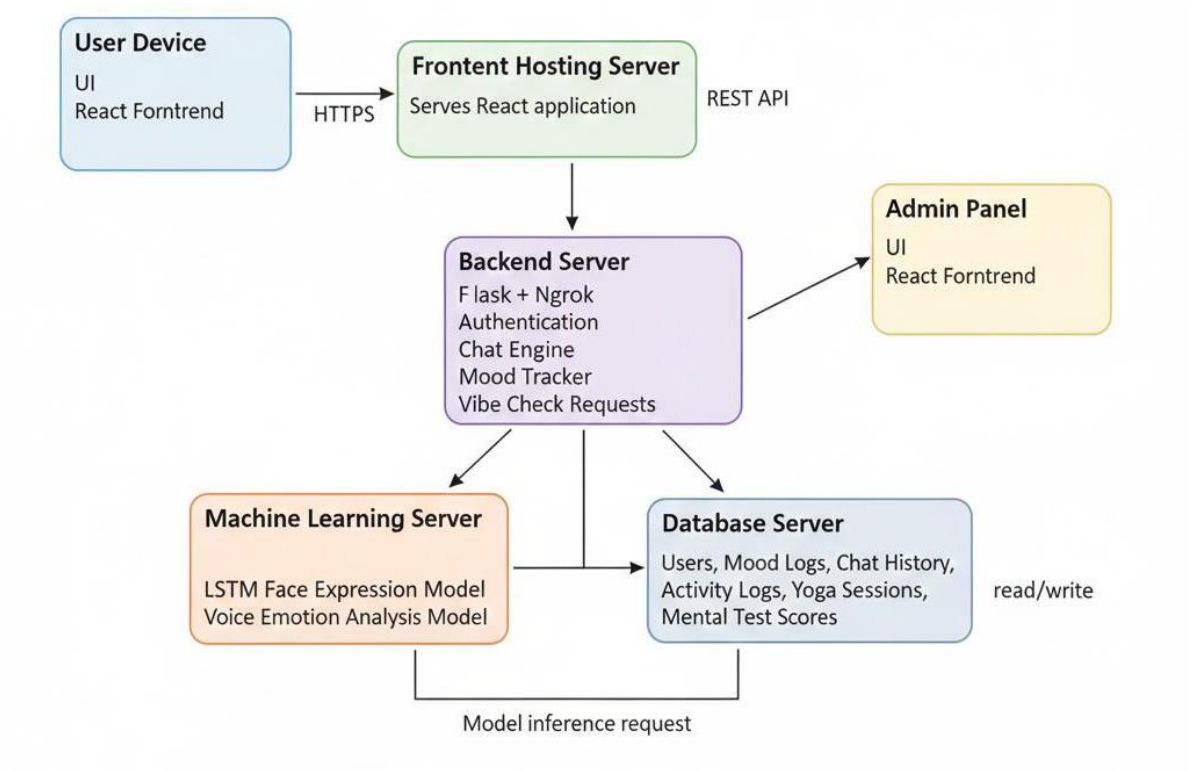

Fig-6: Deployment Diagram

4.3 TECHNOLOGIES USED

Frontend

- React.js – Component-based UI
- HTML5, CSS3, JavaScript
- Chart.js – Visual graphs
- WebRTC – Camera and microphone access

Backend

- Flask (Python) – API development
- Ngrok – Secure tunneling for backend communication
- RESTful API – Data exchange between frontend and backend

Machine Learning / AI

- TensorFlow + Keras – LSTM model for emotion detection
- OpenCV – Face detection and preprocessing
- Librosa– Voice emotion analysis
- NLP Library (Transformers) – VoChat chatbot responses

Database

- MongoDB – Storing user data, scores, logs, journals

Tools & Platforms

- VS Code – Development
- Git – Version control
- Postman – API testing
- Ngrok – Backend exposure during development

CHAPTER-5

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CHAPTER 5: IMPLEMENTATION

The implementation phase focuses on transforming the designed mental health system into a functional and interactive application. This chapter explains how the Mental Health Support System was built using React, Node.js, Flask, and AI models, and how emotional-aware chatbot functionality was integrated with LLAMA AI along with face emotion detection, voice emotion analysis, and sentiment analysis. The implementation also covers dashboard integration, backend logic, and AI service connections to deliver a complete mental wellness platform.

5.1 Setting Up Project Environment & Connections

This step involves preparing the development environment, installing required dependencies, and establishing connections between frontend, backend, AI models, and external APIs such as LLAMA.

5.1.1 Frontend Setup (React.js)

- A React project was created using create-react-app.
- Required libraries such as Axios, React Router, TailwindCSS, and Chart libraries were installed.
- Components were structured for modular development such as:
  - Login/Signup
  - Mood Tracker
  - Gratitude Journal
  - Meditation
  - Yoga Support
  - Emotional Chatbot
  - Dashboard
- Camera and microphone permissions were configured for face and voice emotion detection.
- Environment variables were used to store backend API endpoints.
- Dark mode and responsive UI were implemented for better user experience.

5.1.2 Backend Setup (Node.js & Express.js)

- A Node.js project was initialized with Express.js.

- Server routing and API handling were configured.
- CORS was enabled for frontend and backend communication.
- REST APIs were created for:

/auth
/mood
/gratitude
/goals
/chatbot
/voice
/face
/sentiment
/LLAMA
/dashboard

- Middleware was added for authentication and request validation.
- Node.js acts as the main server and API gateway.

5.1.3 AI Services Setup (Flask)

- Flask was used to host AI models.
- Separate APIs were created for:

/face-emotion
/voice-emotion
/sentiment-analysis
/mental-test

- Python libraries used:
    - TensorFlow
    - OpenCV
    - NumPy
    - Librosa
    - Scikit-learn
    - NLTK
- AI models were deployed and connected to Node.js backend using REST APIs.

5.1.4 LLAMA AI Integration

- LLAMA API was integrated to generate emotional-aware chatbot responses.
- Node.js communicates with LLAMA using API requests.
- User emotion and message context are sent to LLAMA.
- LLAMA generates supportive and empathetic responses.
- Prompt engineering ensures mental health-focused responses.

Example:

User Emotion: Stressed
User Message: I feel anxious
Instruction: Provide supportive and calming response

5.1.5 Environment Configuration

- API keys stored securely in .env files.
- LLAMA API key secured.
- Flask server URLs stored in environment variables.
- Sensitive data not exposed in code.

5.2 Coding the Logic

This section explains how different modules of the mental health system were implemented using frontend, backend, and AI services.

5.2.1 Frontend Logic Implementation

User Authentication

- Login and signup forms created.
- Input validation implemented.
- JWT token stored in local storage.
- Protected routes used.

Mood Tracker

- Users log daily mood.
- Mood data sent to backend.
- Graphs display mental progress.
- Historical data stored and visualized.

Gratitude Journal

- Users write daily gratitude notes.
- Notes saved to backend.
- Displayed in user dashboard.

Meditation and Yoga Modules

- Meditation and yoga sessions listed.
- Users select sessions.
- UI displays guided exercises.

Emotional Chatbot UI

- Chat interface built with React.
- Camera and microphone integrated.
- User message sent to backend.
- Emotion and chatbot responses displayed.
- Suggested mental health activities shown.

5.2.2 Backend Logic Implementation (Node.js)

Authentication Logic

- JWT implemented.
- Password hashing using bcrypt.
- Secure login and signup.
- Protected APIs.

Mood and Gratitude Logic

- Mood entries saved.
- Gratitude entries stored.
- Data retrieved for dashboard.
- Progress tracking enabled.

Emotional Chatbot Logic

- User message received.
- Face data, voice data, and text processed.
- Node.js sends data to Flask services.
- Emotion results collected.
- Emotional state generated.
- Emotional context sent to LLAMA.
- LLAMA response returned to frontend.

Emotion Aggregation Logic

Example:

Face Emotion → Sad
Voice Emotion → Stressed
Text Sentiment → Negative

Final Emotional State:

User is stressed and sad

This is sent to LLAMA for response generation.

5.2.3 AI Model Integration Logic

Face Emotion Model

- Camera captures face.
- Image sent to Flask API.

- CNN/LSTM model detects emotion.
- Emotion returned to Node.js.

Detected emotions:

- Happy
- Sad
- Angry
- Neutral
- Stressed

Voice Emotion Model

- Microphone records voice.
- Audio sent to Flask.
- Librosa extracts features.
- Model detects emotional tone.

Detected emotions:

- Calm
- Anxious
- Angry
- Sad

Sentiment Analysis

- User text processed.
- NLP model analyzes sentiment.
- Output:

Positive
Negative
Neutral

LLAMA Chatbot Integration

- Emotional state combined with user message.
- Prompt created.
- LLAMA generates empathetic response.
- Suggestions added.

Example response:

I understand that you are feeling stressed.
You can try breathing exercises or meditation to relax your mind.

5.3 Connecting the Dashboard

The React dashboard integrates all modules and provides a centralized platform for users.

5.3.1 Dashboard Features

Mood Tracker Dashboard

- Shows mood graph
- Daily tracking
- Mental progress visualization

Emotional Chatbot

- Real-time chat
- Emotion-aware responses
- AI suggestions

Gratitude Journal

- Daily notes
- Mental reflection
- Stored entries

Meditation and Yoga

- Guided sessions
- Relaxation exercises

Mental Test

- Dynamic questions
- Mental score
- Feedback provided

Profile Management

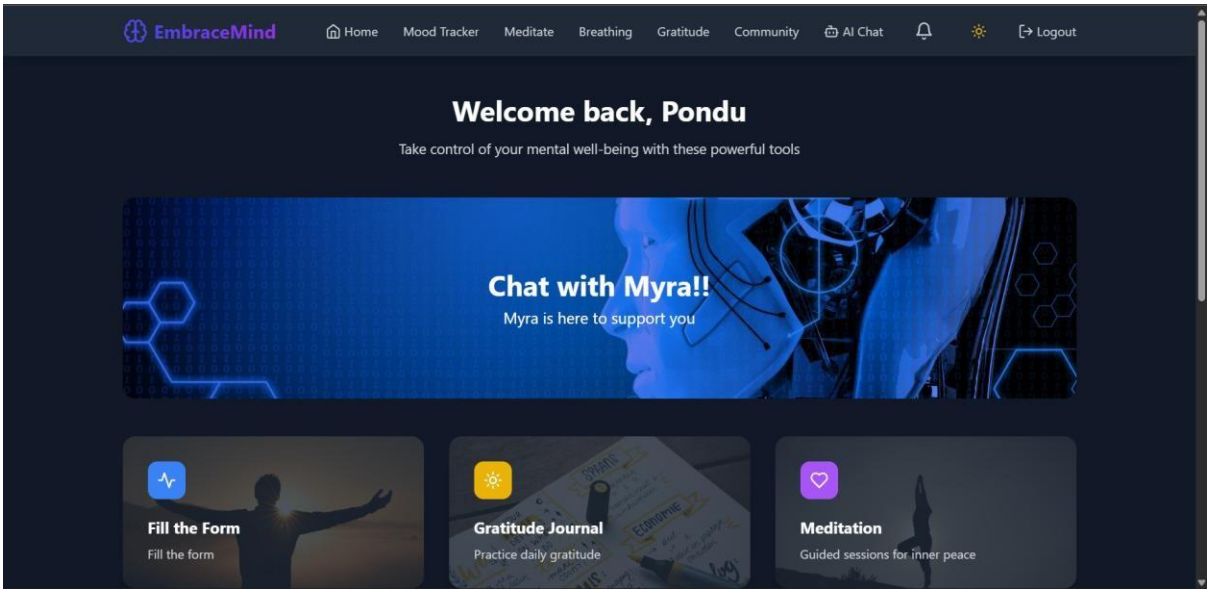
- Update user details
- Password management
- Activity tracking

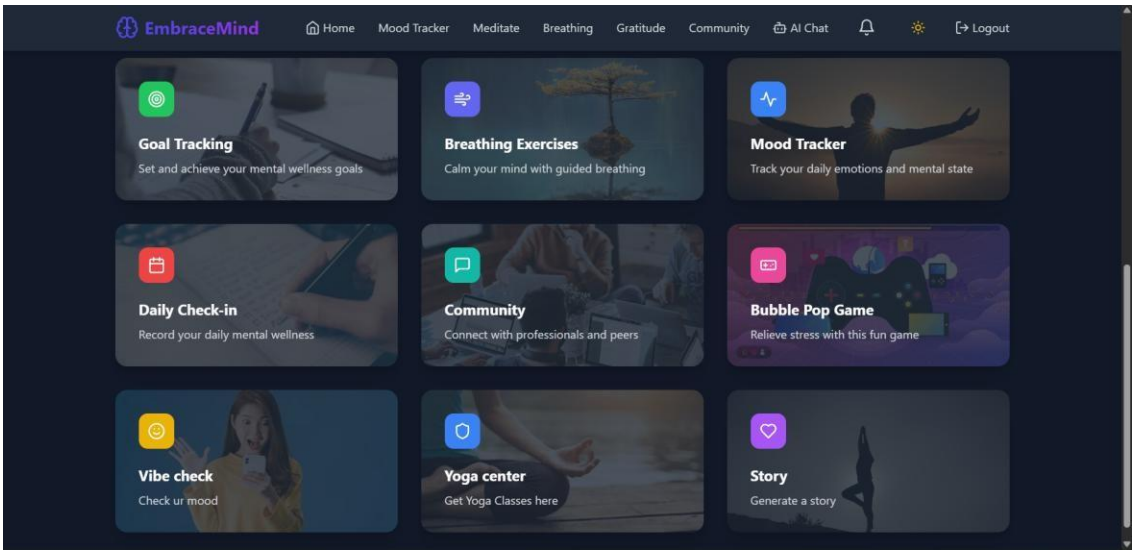
5.3.2 Technical Integration

- React Router used for navigation.
- Axios used for API communication.
- Node.js connects frontend and AI services.
- Flask handles ML inference.
- LLAMA generates intelligent responses.
- TailwindCSS ensures responsive UI.
- State management handles temporary data.
- Dashboard loads user data dynamically.

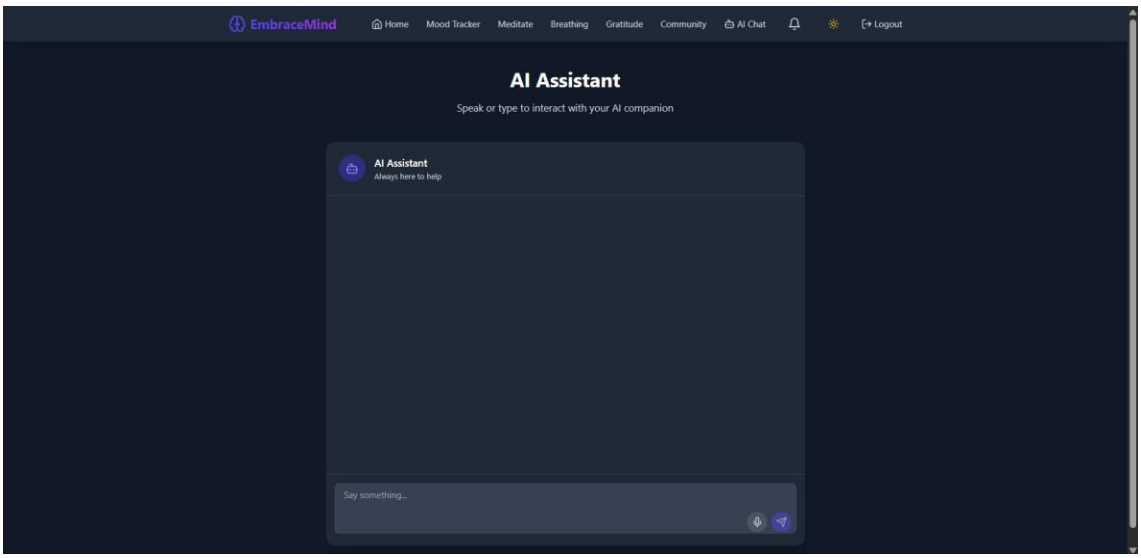
5.4 UI Screenshots

User Dashboard

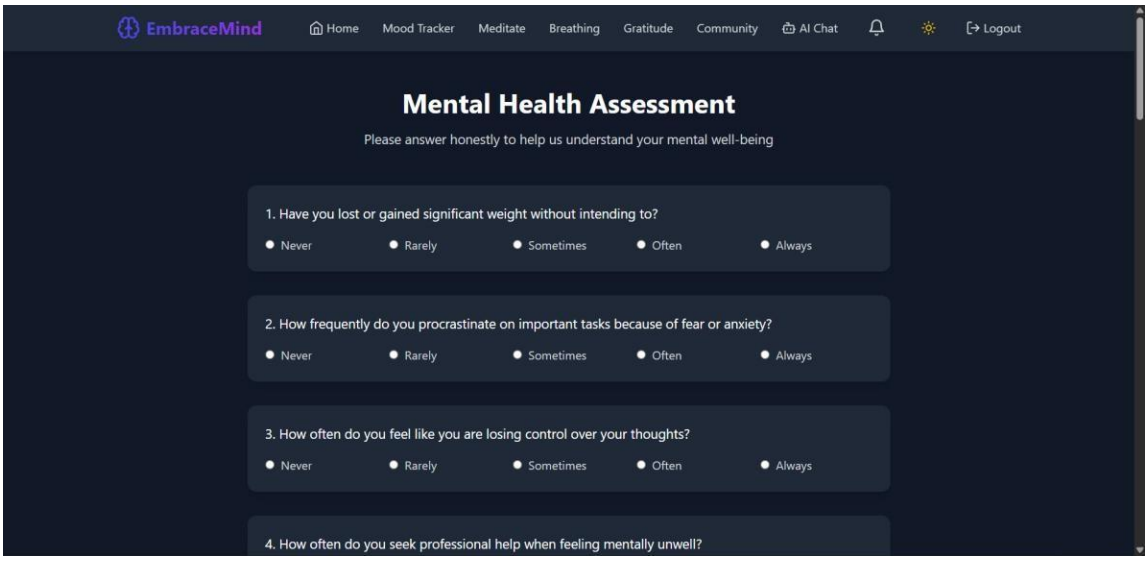




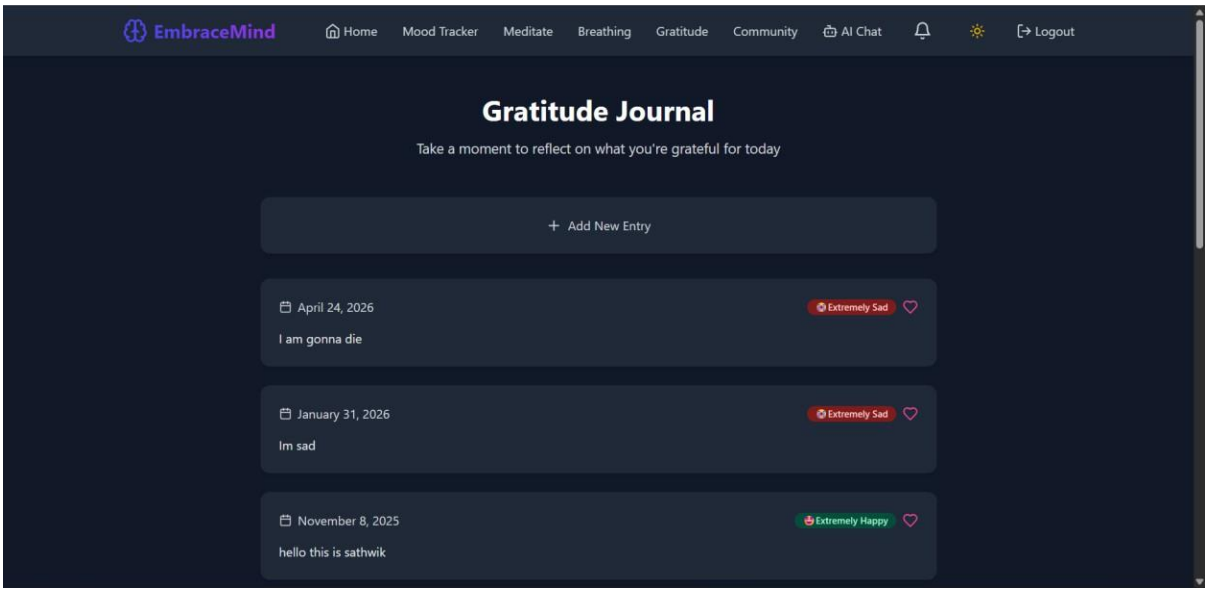
User Chatbot :



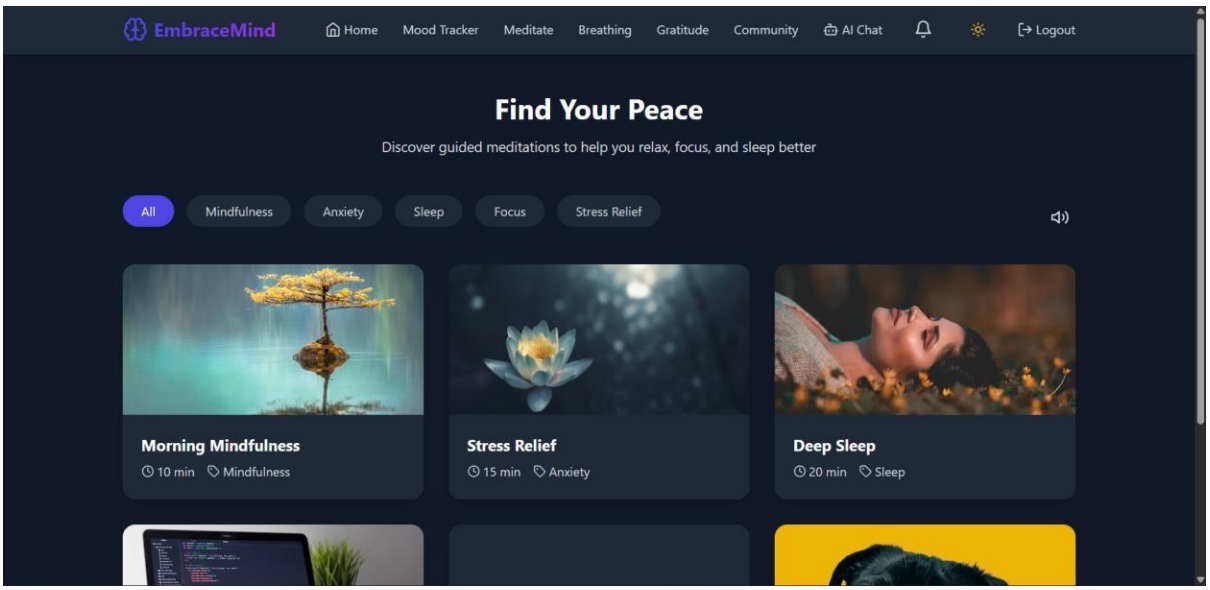
Mental Health Assessment



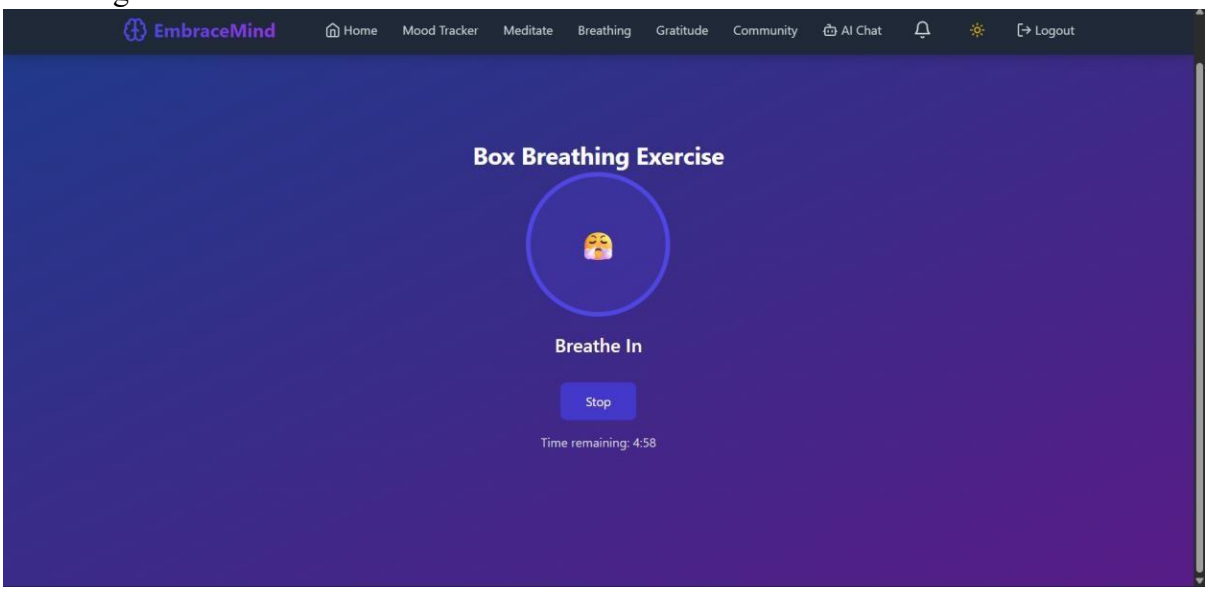
Gratitude Journal



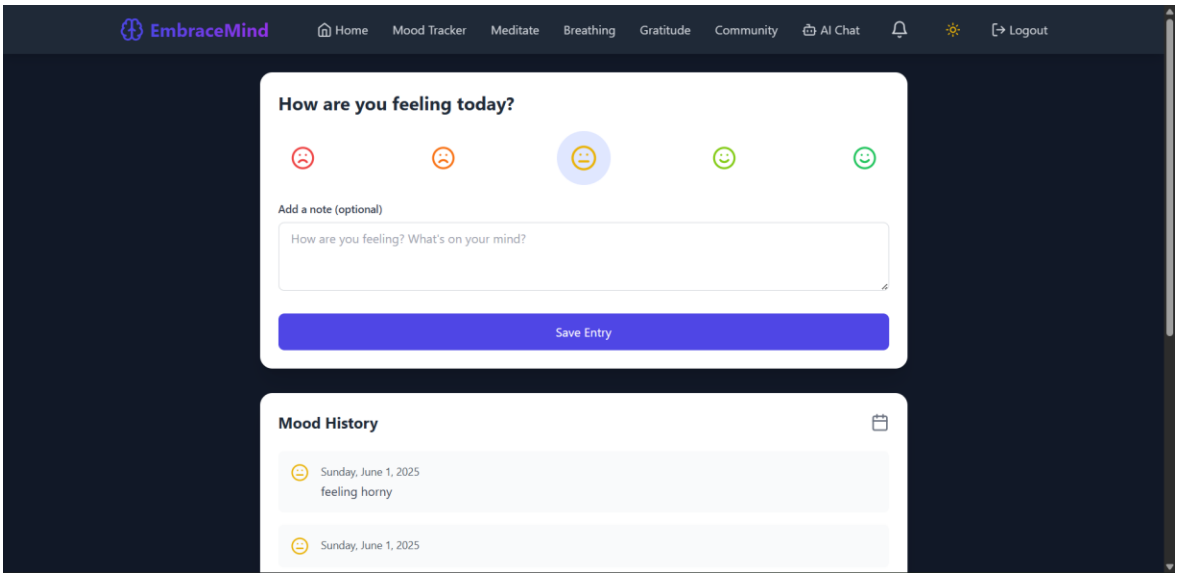
Music Wellness



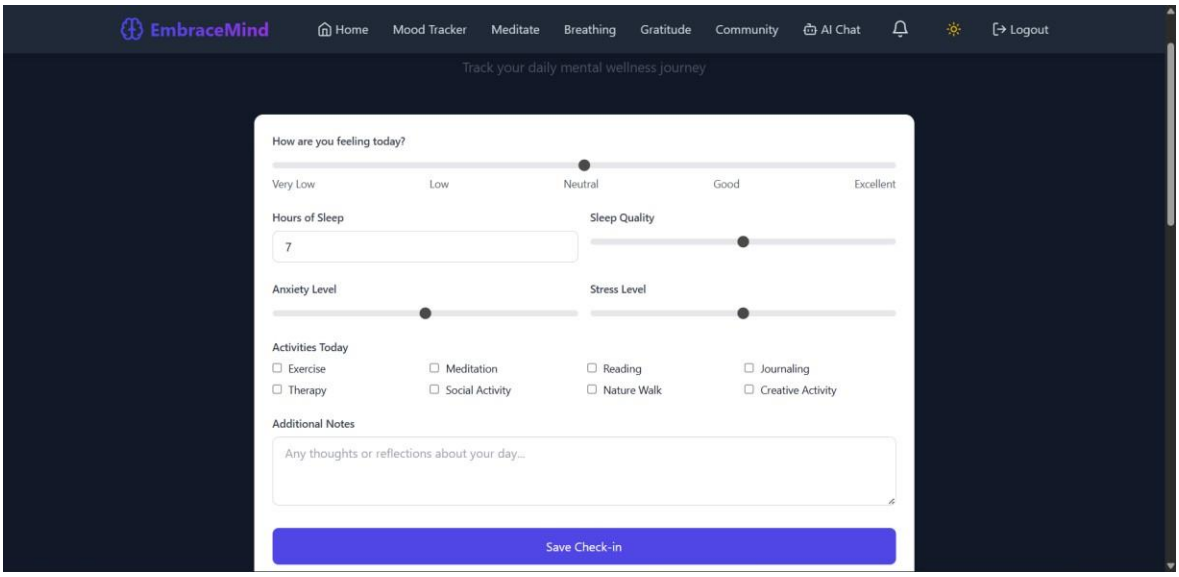
Breathing Exercises



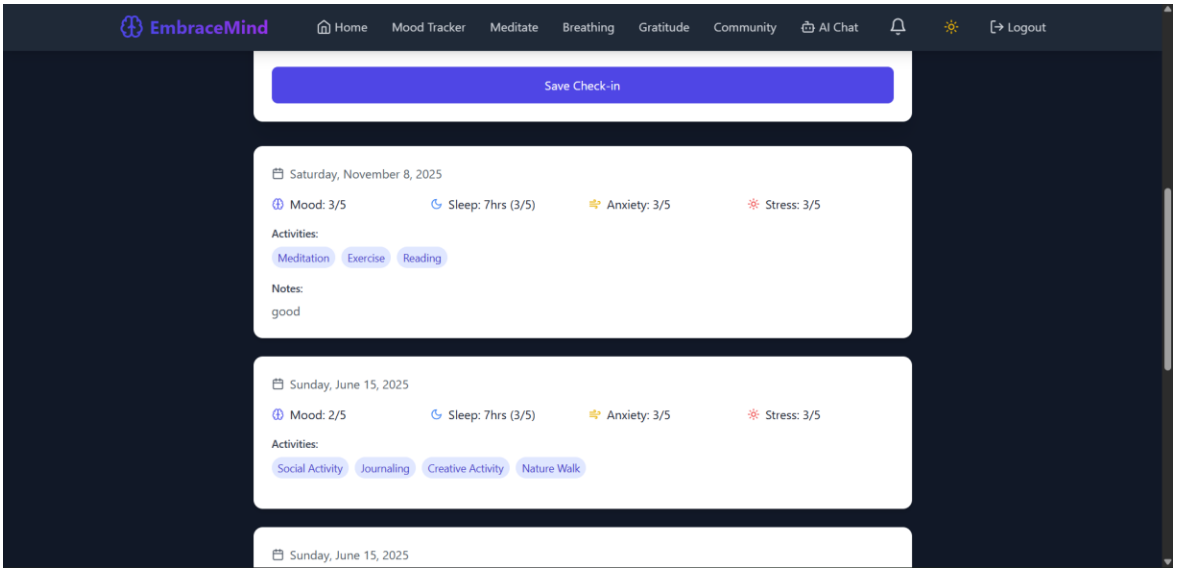
Mood Tracker



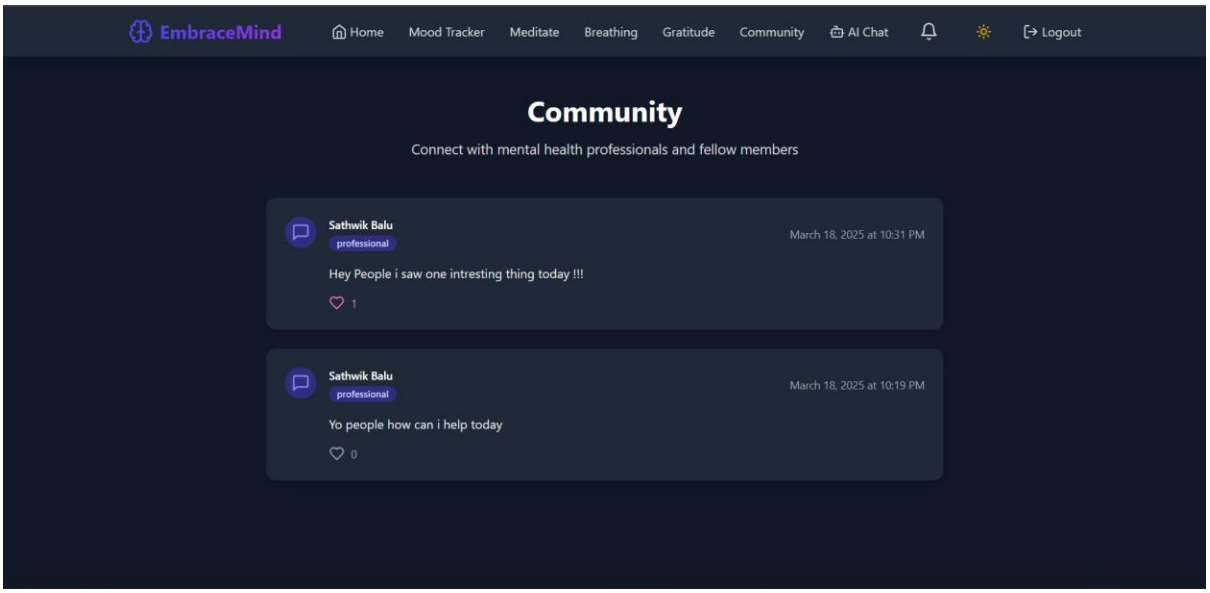
Daily Activities



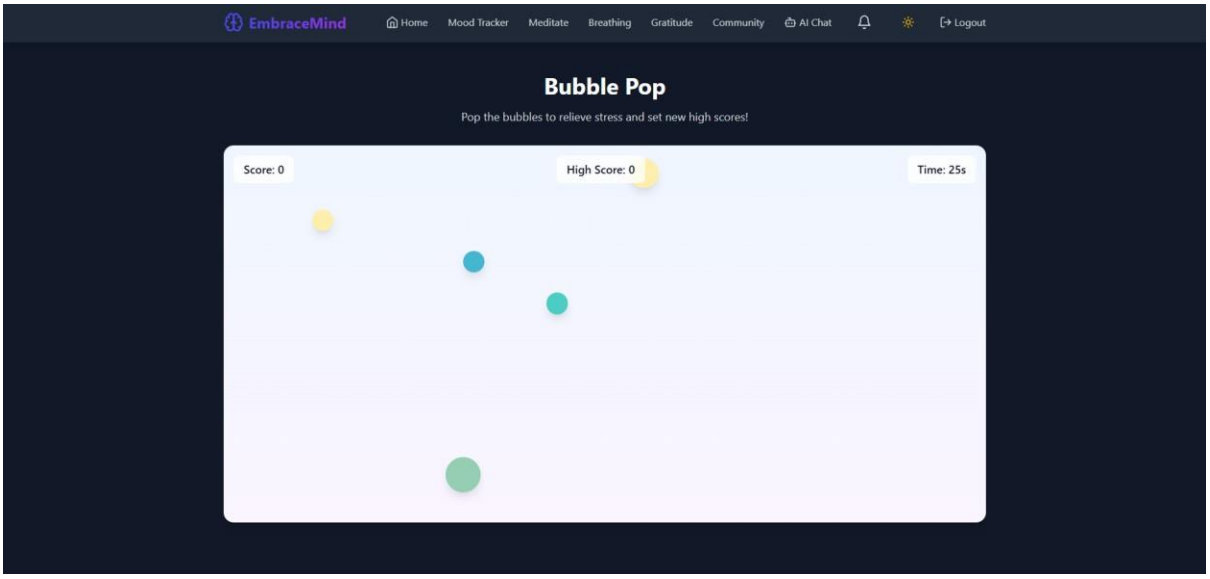
Tracking Daily Activities



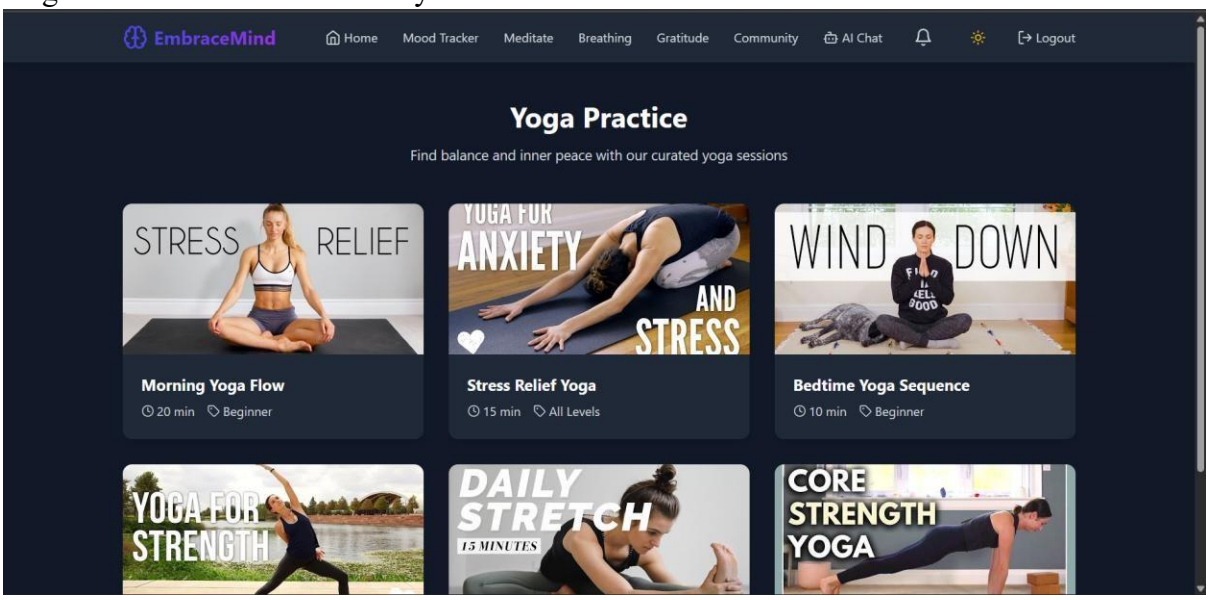
Community



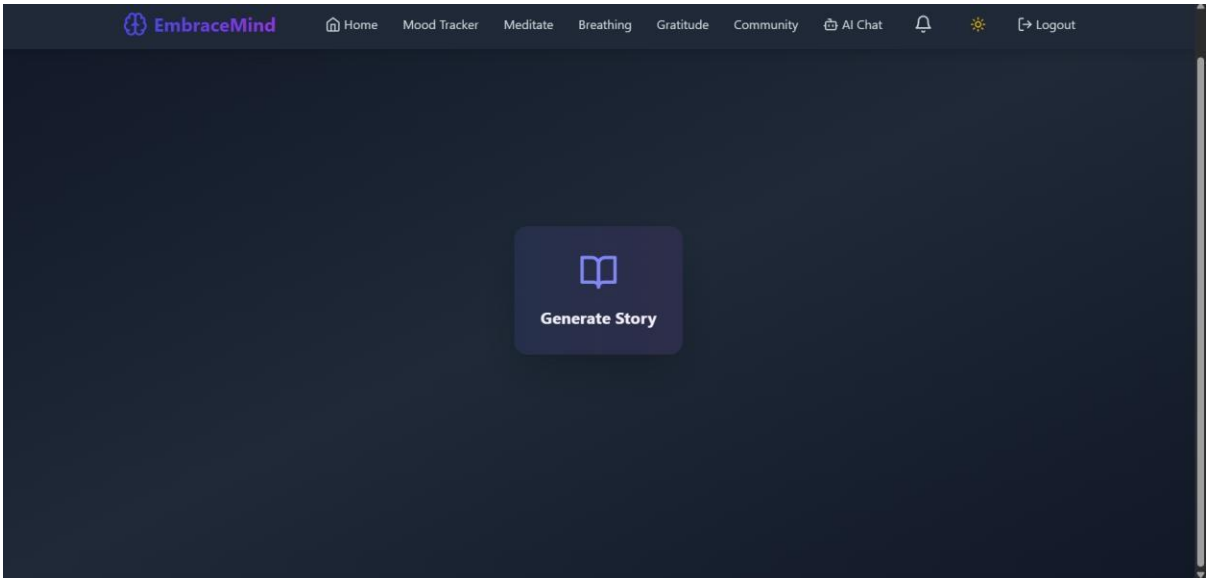
Games to improve mental wellness



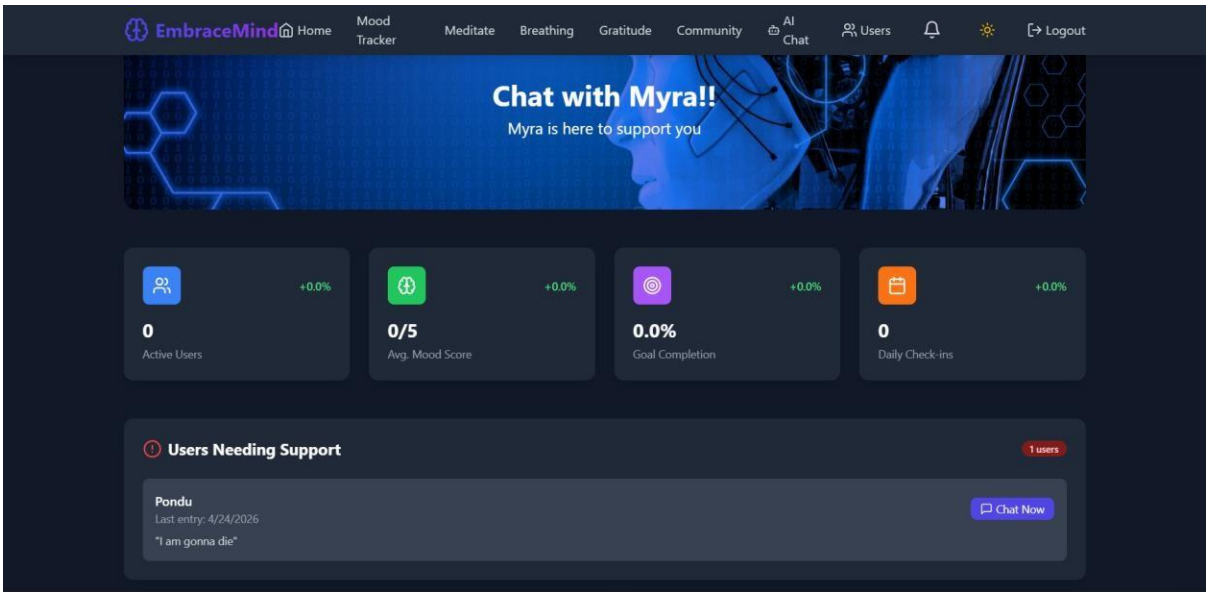
Yoga sessions for mental stability

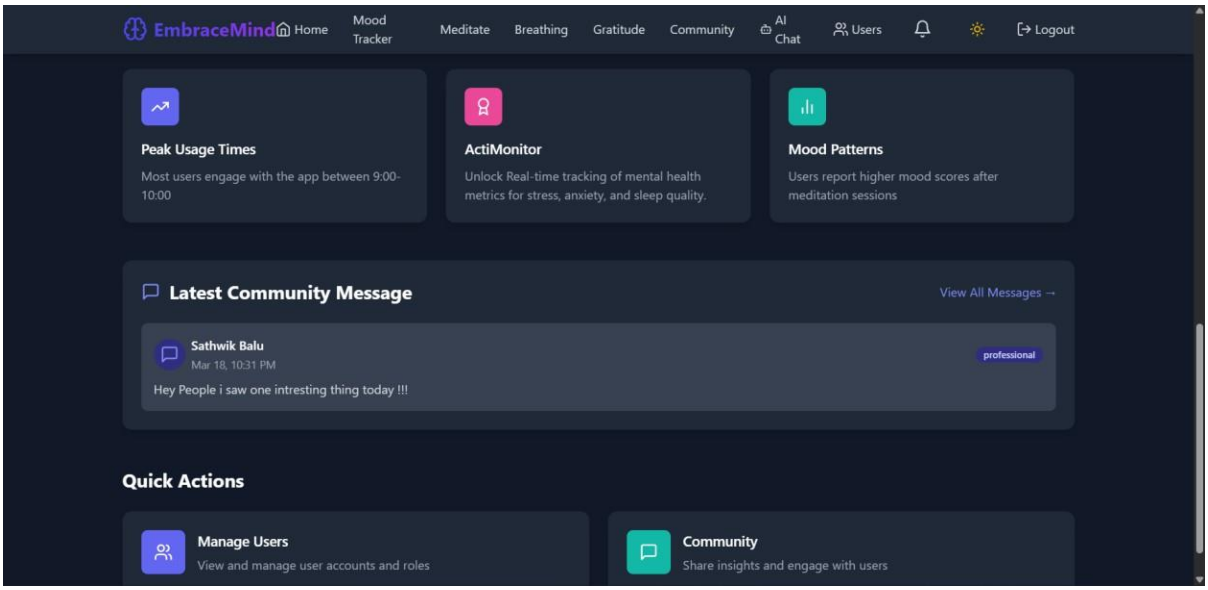


Motivational Stories for users

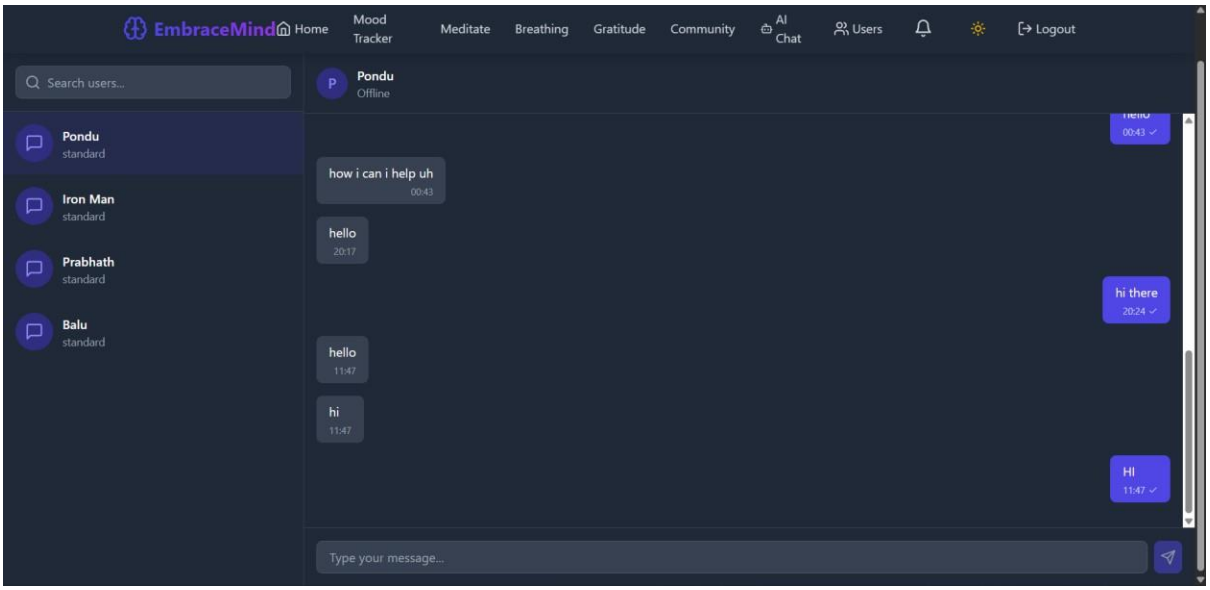


Professional Dashboard

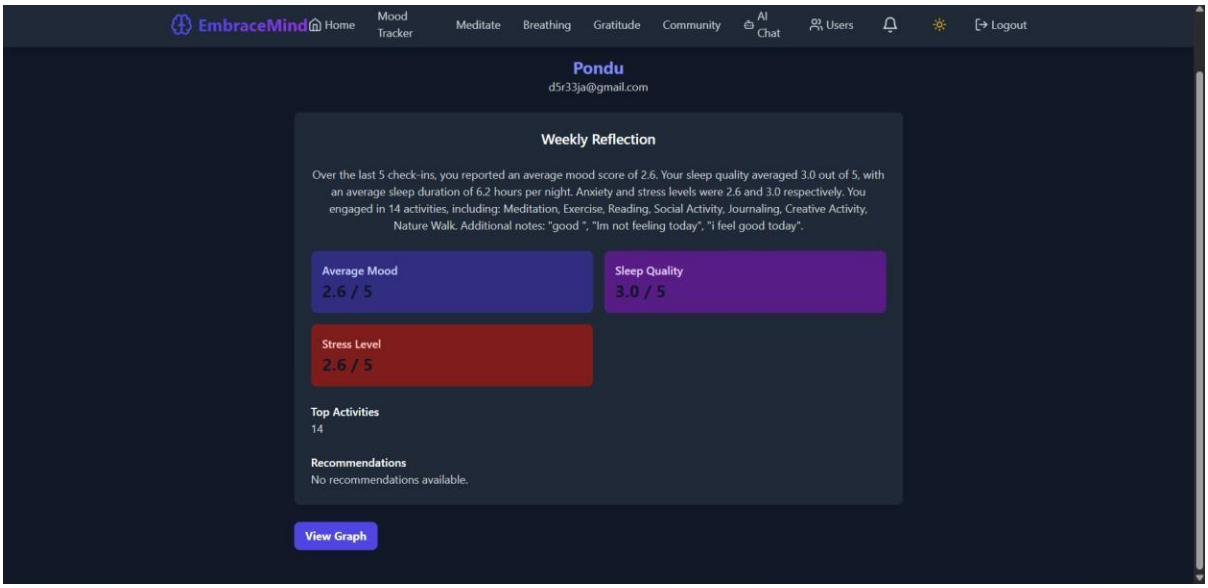




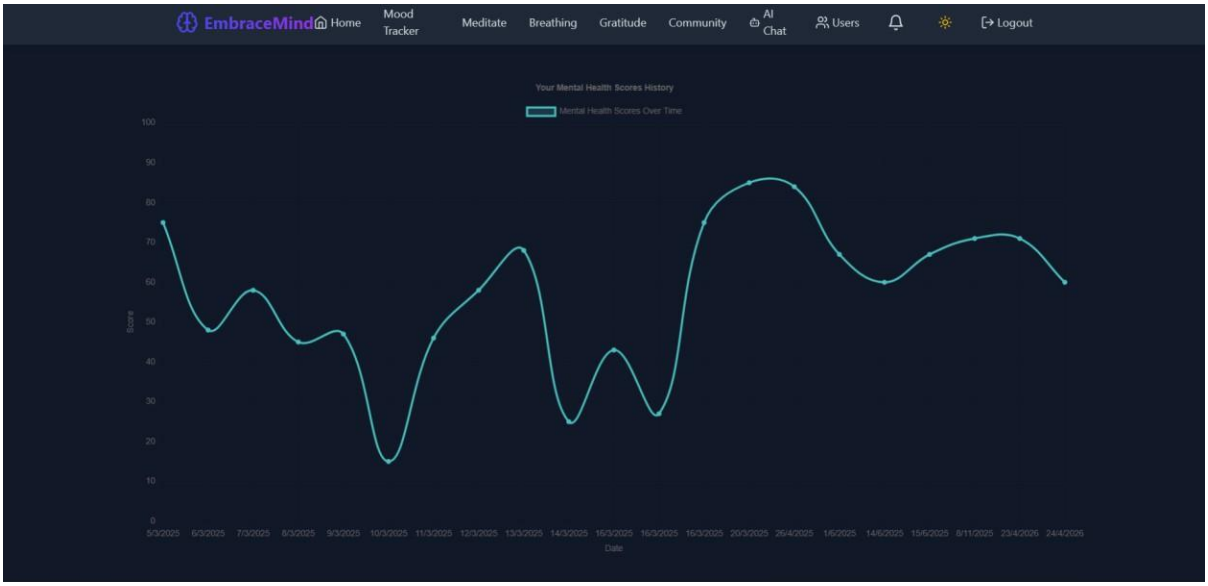
One-to-one chat



Actimonitor for monitoring user weekly activities for professional treatment



Mental Health Tracker



CHAPTER-6

CHAPTER 6: SOFTWARE TESTING

6.1 Introduction

Software testing is a crucial phase in the development lifecycle used to evaluate the functionality, reliability, usability, and performance of the system. It ensures that the developed **Mental Health Application** meets the specified requirements, functions correctly under various conditions, and delivers a smooth and supportive user experience. Testing helps identify defects early, improves system reliability, and ensures stability before deployment, especially since the application deals with sensitive user data and emotional well-being.

6.1.1 Testing Objectives

The main objectives of testing the Mental Health Application are:

1. To ensure functional accuracy
   Verify that all features—including mood tracking, mental health assessment, Vibe Check (face emotion detection), voice modulation analysis, meditation, and chatbot support—work correctly.
2. To validate system reliability
   Ensure the system performs consistently across multiple sessions and user interactions.
3. To ensure usability & user experience
   Confirm that the UI is intuitive, responsive, and easy to navigate for all users.
4. To detect and eliminate bugs
   Identify and fix issues such as incorrect predictions, UI glitches, or backend errors.
5. To verify performance efficiency
   Ensure fast response time for AI predictions, chatbot responses, and data visualization.
6. To confirm compatibility
   Test the system across different browsers, devices, and screen sizes.
7. To ensure secure operation
   Test the system across different browsers, devices, and screen sizes..

6.1.2 Testing Strategies

The following testing strategies were adopted:

1. Unit Testing

- Individual modules such as mood tracking, AI prediction models, chatbot, and frontend components were tested separately.
- Ensured logical correctness at the code level.

2. Integration Testing

- Verified that integrated modules (Frontend ↔ Backend ↔ AI Models ↔ Database) worked seamlessly.
- Checked API calls, data flow, and response correctness.

3. System Testing

- The entire system was tested as a whole to verify compliance with SRS.
- Included testing complete workflows like:

  User login → Mood input → AI analysis → Feedback → Graph visualization

4. Functional Testing

- Verified all functionalities such as emotion detection, journaling, chatbot interaction, and daily tracking.

5. Non-Functional Testing

- Evaluated system performance, scalability, and reliability.
- Tested response time for AI features and dashboard loading.

6. User Acceptance Testing (UAT)

- Conducted testing with real users.
- Feedback collected on usability, comfort, and usefulness of mental health features.

6.1.3 System Evaluation

The Mental Health Application was evaluated based on:

1. Functional Completeness:
   All major features such as mood tracking, chatbot, emotion detection, and journaling work effectively.
2. Accuracy of AI Output:
   Evaluated correctness of emotion detection, voice analysis, and personalized feedback.
3. Performance & Load Handling:
   Tested system under multiple user requests and ensured smooth performance.
4. User Experience:
   The interface is simple, responsive, and supports easy interaction.
5. Stability:
   The system runs without crashes during continuous usage.

6.1.4 Testing New System

The new system was tested thoroughly before being deployed. Key aspects tested:

- New features:
  Vibe Check (face emotion detection), voice analysis, mood tracking, AI chatbot, and mental health scoring.
- AI model integration:
  Smooth functioning of emotion detection models and AI-based predictions.
- Database operations:
  Efficient storage and retrieval of user data such as mood logs, journal entries, and progress tracking.
- Cross-Platform compatibility:
  Tested across web browsers and devices (desktop and mobile).
- Error handling:
  Proper error messages for invalid inputs, failed predictions, or connectivity issues.

6.2 Test Cases

Below are sample test cases included in the testing phase:

TC ID	Test Case Name	Description	Input	Expected Output	Actual Output	Status
TC_01	Mood Tracking Test	Verify mood logging functionality	User selects mood	Mood saved and displayed in dashboard	Mood saved successfully	Successful
TC_02	Emotion Detection Test	Test Vibe Check feature	User face input	Correct emotion detected	Emotion detected accurately	Successful
TC_03	Voice Analysis Test	Check voice modulation detection	User voice input	Emotional state identified	Correct analysis shown	Successful
TC_04	Chatbot Response Test	Verify AI chatbot interaction	User message	Relevant response generated	Response generated correctly	Successful
TC_05	Input Validation Test	Check invalid/missing inputs	Empty fields	Error message displayed	Error handled properly	Successful
TC_06	Performance Test	Test system under load	Multiple user requests	Fast response time	Response within limits	Successful

TC ID	Test Case Name	Description	Input	Expected Output	Actual Output	Status
TC_07	Data Storage Test	Verify saving user data	Mood/journal entry	Data stored correctly	Data retrieved successfully	Successful
TC_08	UI Navigation Test	Test navigation between pages	User clicks pages	Smooth navigation	No issues observed	Successful
TC_09	Compatibility Test	Test across browsers	Chrome, Firefox, Edge	Consistent behavior	Same across platforms	Successful
TC_10	Security Test	Ensure data safety	User data input	Secure storage and access	No vulnerabilities found	Successful

CONCLUSION

The Mental Health Application represents a significant advancement in the integration of Artificial Intelligence with healthcare and well-being support systems. Traditional mental health solutions often lack personalization and real-time interaction, but this project overcomes those limitations by providing an intelligent, user-centered platform that adapts to individual emotional and psychological needs.

By incorporating advanced AI techniques such as facial emotion recognition (Vibe Check), voice modulation analysis, and dynamic mental health assessment, the system offers accurate insights into a user's emotional state. The integration of features like mood tracking, meditation, breathing exercises, gratitude journaling, and AI-based chatbot support creates a holistic ecosystem for mental wellness. These tools not only help users monitor their mental health but also actively guide them toward improvement.

The use of modern web technologies such as React and Flask ensures a scalable, responsive, and efficient platform. The system architecture enables seamless interaction between user inputs, AI processing, and real-time feedback, delivering a smooth and engaging user experience. The application emphasizes data privacy and ethical handling of sensitive user information, which is critical in mental health systems.

Furthermore, the project highlights the growing importance of personalization in digital healthcare. By offering tailored recommendations, emotional support, and progress tracking, the system empowers users to take control of their mental well-being. It serves not only as a support tool but also as a preventive solution by identifying emotional patterns early.

Overall, the Mental Health Application successfully achieves its objectives by combining AI innovation with user-centric design. It demonstrates the potential of technology in addressing real-world mental health challenges and lays a strong foundation for future enhancements and large-scale deployment. As technology evolves, such systems can play a vital role in making mental health support more accessible, proactive, and effective.

FUTURE ENHANCEMENTS

To further improve the system, the following enhancements can be implemented:

1. **Multilingual Support**

   Enable support for multiple languages to make the application accessible to a broader and diverse user base.

2. **Voice-Based Interaction**

   Integrate speech-to-text and text-to-speech features for hands-free interaction and guided mental wellness sessions.

3. **Advanced Emotion Detection**

   Enhance accuracy using deep learning models for better facial and voice-based emotional analysis.

4. **Wearable Device Integration**

   Connect with smartwatches and fitness bands to monitor stress levels, heart rate, and sleep patterns.

5. **Personalized Therapy Recommendations**

   Provide AI-driven suggestions such as therapy sessions, exercises, or professional help based on user condition.

6. **Community Support Platform**

   Introduce peer-to-peer support forums for sharing experiences and encouragement.

7. **Mobile Application Development**

   Develop Android and iOS apps for improved accessibility and engagement.

8. **Gamification Features**

   Add rewards, streaks, and interactive games (like Bubble Pop) to encourage consistent usage.

9. **Cloud Integration & Analytics**

   Implement secure cloud storage and advanced analytics for tracking long-term mental health progress.

   .

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